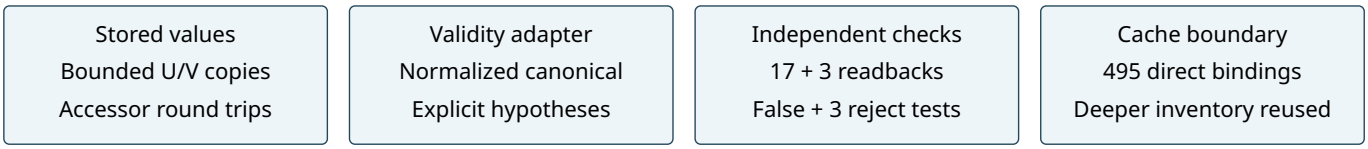
Independent fixed-array constructor review

Independent replay and source inspection recorded passed, faithful_scoped and validated_scoped for ten modeled declarations and seven producer controls at commit e62d035ab852be5bbfd27bbe8f11c8f6bfe25f2e. The prospective divisor-array-copy review plan was published in that commit. This archive preserves the completed review without rerunning proofs or changing research status.



Construction checked. Native three_leads.rs lines 943-964 initializes the identity, assigns degree len(U)-1, zeros U, then copies U and V into capacities three and two. The model preserves those values and copy order. Proved bounds ensure every modeled write is in range. The support accessor returns U; the trimmed ordinate accessor returns V under its explicit normalization assumption. Decoded monicity and degree bounds yield shape for normalized canonical inputs. Decoded validity is an assumption transported through storage, not inferred merely from shape.

Replay record. Five target modules compiled into a fresh private directory. Seventeen producer named axiom readbacks and three reviewer controls/readbacks passed. The false theorem was rejected with exit 1. Empty inventory, injected sorryAx and wrong declaration-name controls were rejected by the Rust checker. No unexpected replay failure occurred. The nonmonic-admitted reviewer control separates constructor acceptance from validity; trailing-support rejection and maximum padding controls separate shape from normalization.

Exact cache boundary. Target caches were fresh. The Lean binary and 495 direct helper source/cache files were hashed before and checked after. Deeper dependency unions were checked against a retained earlier inventory during review; they were not a fresh pre-execution snapshot. Imported caches remained reused. There was no independent binary/kernel rechecker and no proof excluding transient concurrent edits. This is source inspection and replay of named declarations, not a rebuild of all imports. The non-author reviewer inherited task history and consulted no sibling verdict.

Scope and retained observations. The F17 control supplies existential stored arrays from earlier existential outputs, not a concrete native trajectory. Raw copy/accessor results use their stated shape and normalization assumptions; canonical padding requires $p>0$; the validity adapter uses prime p and $1<p$. Option.none models assertion rejection, not panic semantics. Physical copying, usize/compiler behavior, allocation, event counters, Picard classes and whole addition dispatch remain outside this component. The initial local tracking ref 27b3b840 was stale because of narrow fetch configuration. After root refreshed it, main 66aeee272c250868d059f1cd27165e9805641048 had byte-identical native source and no target-packet diff; no replay was required. Both old and refreshed logs and the rules diff are retained. The initial session-log template selection and correction also remain recorded.

Evidence and next joint. review/REVIEW.md and review/review.json record the claim-by-claim result, sources read and independence attestation; review/input, metadata and logs preserve source bytes, bindings and checker results. Producer evidence remains in divisor-array-copy. No scientific cohort, timing, cost ratio or campaign limit changed. The diagram shows proof/review structure because this packet contains no comparative measurement. Next is whole-dispatch composition with stored input extraction. Canonical scientific graphs remain unchanged.